192.168.1.0 — 192.168.1.255

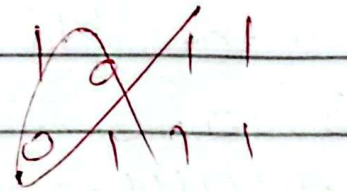
4 octets IPv4

~~0-126~~

0-127	Class A	N 255.0.0.0	H	H H
128-191	Class B	N 255.255.0.0	N	H H
192-223	Class C	N 255.255.255	N	N H

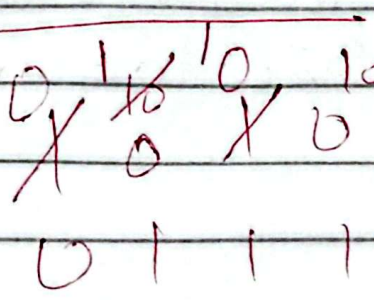
Class A

255.0.0.0



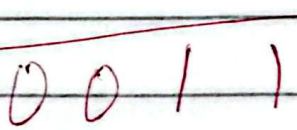
Class B

255.255.0.0



Class C

255.255.255.0



Example #01

Starting Network: 172.168.0.0

$2^n \geq$ number of subnets needed

$2^n \geq 60$

$2^6 \geq 60$

172.168.0.0/16

255.255.0.0

11111111.11111111.00000000.00000000

32 - 16 = 16 host bits

IP Address: 100.10.20.0

100 host per subnet
30 subnets recquired

Step 1

Class A (100.10.20.0
↓
class A

Host Needed: 100 → Upper Bound
$= 2^7 \geq 100$

128 - 2 = 126
Usable

Upper bound: $128 \geq 100$

$2^5 \geq 30$

$32 \geq 30$

Subnet 255.000.000.000

11111111.00000000.00000000.00000000

Step 3

Total required: $2^7 \times 2^5 = 4096$

Class A capacity = 2^{24} = 16 Million

4096 << 16M

possible

Step 4

11111111 . 11111000 . 00000000 . 00000000

255 . 248 . 0 . 0

Step 5:

256 - 248 = 8

100 . 0 . 0 . 0

LAB #09:-

221-2327 M. Rohaan Sec-E

Example #01:-

Ip-address: 172.168.1.0
No. of recquired hosts: 160 /subnet
No. of recquired subnets: 60 subnets

Step #01:-

172.168.1.0

↓

Class B

Host Needed (upper Bound)

$2^8 \geq 160$

$256 \geq 160$

Subnets Needed (Upper Bound)

$2^6 \geq 60$

$64 \geq 60$

Subnet of the class of ip address

255.255.0.0

Step # 02:-

255 . 255 . 0 . 0

11111111 . 11111111 . 00000000 . 00000000

Step # 03:-

Total Required : $2^8 \cdot 2^6 =$ 16384

Class B capacity : $2^{16} = 65536$

16384 << 65536

Subnetting possible

Step # 04:-

Upper Bound 2^6, (Borrow 6 bits from Network)

11111111 . 11111111 . 11111100 . 00000000

255 . 255 . 252 . 0

Step #05:

256 - 252 = 4

Subnetting by each block size of 4.

Step #06:-

First Subnet:

172.168.0.0 - 172.168.3.25

range is 4 (0, 1, 2, 3)

Network address :- 172.168.0.0

Broadcast address :- 172.168.3.255

Second Subnet:

172.168.4.0 - 172.168.7.255

NA :- 172.168.4.0

BA :- 172.168.7.255

CamScanner

Third Subnet:

172.168.8.0 – 172.168.11.255

NA: 172.168.8.0

BA: 172.168.11.255

Fourth Subnet:

172.168.12.0 – 172.168.15.255

NA: 172.168.12.0

BA: 172.168.15.255

Subnet #	NA	FU	LU	BA
1	172.168.0.0	172.168.0.1	172.168.2.254	172.168.3.255
2	172.168.4.0	172.168.4.1	172.168.7.254	172.168.7.255
3	172.168.8.0	172.168.8.1	172.168.11.254	172.168.11.255
4	172.168.12.0	172.168.12.1	172.168.15.254	172.168.15.255

Hosts: $2^{10} = 1024$

Example #02:

IP address: 10.10.20.0

No. of required hosts: 100 host/subnet
No. of required subnets: 100 subnets

Step#01:

10.10.20.0
↓
Class A

Host Needed (Upper Bound)

$2^7 \geq 100 \Rightarrow 128 \geq 100$

$2^7 \geq 100 \Rightarrow 128 \geq 100$

Step#02:

Subnet of the class of Ip address

255.0.0.0

11111111.00000000.00000000.00000000

Step # 03:-

Total Required: $2^7 \cdot 2^7 = 16384$

Class A capacity: 16777216

$16384 << 16777216$

Subnetting is possible

Step # 04:

Upper Bound 2^7

(Borrow 7 bits)

11111111.1111110.00000000.00000000

255.254.0.0

Step # 05:-

256 - 254 = 2

Blocks of 2 in each subnet.

Step # 06:

First Subnet:

NA BA

10.0.0.0 - 10.1.255.255

Second Subnet:

NA BA

10.2.0.0 - 10.3.255.255

Third Subnet:

NA BA

10.4.0.0 - 10.5.255.255

Fourth Subnet:

NA BA

10.6.0.0 - 10.7.255.255

No Usable Address

Example #03:-

IP-address:- 192.168.29.1
No. of required hosts:- 100 host/subnet
No. of required subnets:- 60 subnets

Step #01:-

192.168.29.1
↓
Class C
No. of Host Need:- (Upper Bound)

$2^7 \geq 100$
$2^6 \geq 60$

Step #02:-
Subnet of the class of IP address

255.255.255.0
11111111.11111111.11111111.00000000

Step # 03:

Total Required = $2^7 \cdot 2^6 = 8192$

Class C capacity = $2^8 = 256$

$8192 >> 256$

Subnetting is not possible.

CamScanner

Example #04:

IP address:- 220.162.49.1

No. of required hosts:- 30 host/subnet
No. of required subnets:- 60 subnets

Step # 01:

220.162.49.1
↓
Class C

No. of Host & Subnets Need (Upper Bound)

$2^5 \geq 30$

$2^6 \geq 60$

Step #02:

Subnet of the class of IP address

255.255.255.0

11111111.11111111.11111111.00000000

Step #03:-

Total Required:- $2^5 \cdot 2^6 = 2048$

Class C capacity:- $2^8 = 256$.

>> 256

Subnetting is not possible